Probabilistic System Identification and State Estimation via Normalizing Flows

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I dedicate this thesis to my family.

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Acronyms

DGM Deep Generative Model

VAE Variational Autoencoder

NF Normalizing Flow

BNN Bayesian Neural Network

RNN Recurrent Neural Network

NARX Nonlinear Autoregressive eXogenous

MPC Model Predictive Control

DGP Deep Gaussian Process

PDF Probability Density Function

MCMC Markov Chain Monte Carlo

MH Metropolis-Hastings

MALA Metropolis-adjusted Langevin algorithm

HMC Hamiltonian Monte Carlo

ELBO Evidence Lower Bound

KL Kullback-Leibler

VI Variational Inference

GPR Gaussian process regression

SE squared exponential

RBF Radial Basis Function

DGP Deep Gaussian Process

SGD Stochastic Gradient Descent

BbB Bayes by Backprop

CNN Convolutional Neural Network

RNN Recurrent Neural Network

SDE stochastic differential equation

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Chapter I

Introduction

1 Motivation

In this seemingly certain world, uncertainty lingers like a shadow, yet from the uncertain, we often find certainty.

The paradoxical relationship between certainty and uncertainty manifests across disciplines, demonstrating how truth often arises from navigating ambiguity. Quantum mechanics redefines predictability under randomness in the subatomic world, where Newtonian deterministic physics collapses. This uncertainty led to groundbreaking laws (e.g. Heisenberg's Uncertainty Principle), which now underpin modern technologies like quantum computing and all forms of spectroscopy. An example of medical diagnosis is that a patient presents vague symptoms, such as fatigue or pain, which could indicate anything from stress to cancer. Doctors converge on the cause through iterative hypothesis-testing approaches. Climate science also epitomizes how we manage uncertainty. Climate models incorporate a wide range of sources of variability and uncertainty. A common way to address it is to combine the results of a collection of different simulation models. By running multi-model ensembles, scientists can identify consistent patterns across the uncertainty.

From quantum randomness to medical diagnosis to climate models, uncertainty does not destroy the deterministic systems, but provides us accessibility to identify the systems. This perspective supports the use of ***probabilistic inference*** methods, which offer a rigorous framework for representing uncertainty and reasoning from uncertainty. In statistics, modeling uncertainty of variables from given data samples, using a probability distribution rather than just a point estimate is known as inference. The estimated distribution facilitates uncertainty quantification and provides a measure of reliability of the results, making it particularly well-suited for real-world applications.

The mathematical foundation of probabilistic inference is Bayes' rule: beliefs about unknown variables (the prior) are updated using observed data (likelihood), yielding the posterior distribution. A simplified formulation is:

$$\text{posterior} = \text{prior} \times \text{likelihood} \;/\; \text{total evidence or marginal likelihood.} \tag{I.1}$$

For example, we can illustrate Bayes' rule using the medical diagnosis scenario: the objective is to obtain the probability of a disease given a positive test result:

$$P(\text{Disease}|\text{Positive}) = \frac{P(\text{Positive}|\text{Disease}) \times P(\text{Disease})}{P(\text{Positive})},$$
$$P(\text{Positive}) = P(\text{Positive}|\text{Disease}) \times P(\text{Disease}) + P(\text{Positive}|\overline{\text{Disease}}) \times P(\overline{\text{Disease}}),$$

where $\overline{\text{Disease}}$ denotes "No Disease". In the Bayesian context, probabilistic inference is the process of obtaining the posterior distribution of target variables from observed data. This perspective extends naturally to modern machine learning, where predictions, structures, and policies can all be understood as random variables and probability distributions (Murphy, 2022).

Especially in modern Deep Generative Models (DGMs), probabilistic inference plays an essential role (Murphy, 2023). Through introducing random variables to represent complex data in latent space, DGMs persue data sampling from latent variables (generation) based on the recognition of underlying latent representation from data (inference). However, inference of the posterior over the latent variables directly using Bayes' rule I.3 is intractable, because all possible values of the latent variables are rarely obtainable in complex and high-dimensional models. To fully unfold the potential of DGMs, this issue has to be at first addressed. Variational Autoencoders (VAEs) (Kingma and Welling, 2019) tackle this problem by introducing an inference network as a tractable approximation to the posterior. Normalizing Flows (NFs) (Rezende and Mohamed, 2016) construct bijective transformations between observed data and latent variables, making exact computation of posteriors possible. Diffusion models (Ho et al., 2020) avoid calculating posterior over latent variables by designing it as a simple Gaussian distribution, framing both inference and generation as iterative denoising from noise via learned transformations.

To sum up, DGMs are inherently probabilistic; through probabilistic modeling and inference, they are powerful in extracting latent information and representing uncertainty in complex data. These capabilities are proven in numerous applications in images, text, and speech. The same principles extend naturally to scientific and engineering domains. One prominent example is: how can probabilistic inference in DGMs empower the identification and modeling of complex, stochastic, nonlinear dynamical systems? In control theory, the process of building mathematical models of dynamical systems from observed input-output data is called ***system identification***. Interpreting system identification in probabilistic terms, dynamics identification is inferring the posterior of the next states based on the previous states and system inputs, while parameter identification is estimating the posterior over system parameters given observations.

Classic methods for system identification rely on linearized models or pre-specified

nonlinear structures. Model parameters are then estimated from observed data using estimation techniques such as the least squares method and Bayesian estimation. Subsequently, a validation process is employed to assess and select the most appropriate model structure. Although these methods offer strong physical interpretability and are often data-efficient, they exhibit several limitations: parameter estimation can be computationally inefficient with large amounts of data; results are sensitive to the choice of model structures and dependent on expert knowledge; uncertainty is difficult to handle and quantify; and they are generally inapplicable to partially observable systems and high-dimensional and raw data.

Despite the demand for uncertainty quantification, deep learning methods are suitable for system identification thanks to their strength in representing information from complex data, flexibility of function approximation, scalability in high-dimensional data, and compatibility with optimization techniques. Taking uncertainty quantification into account, a simple yet effective technique is Gaussian Process Regression (Särkkä, 2019), which involves assigning a Gaussian process prior to the latent regressor and performing statistical conditioning based on the available observations. This method has gained widespread adoption due to its usability and adaptability. Nonetheless, its significant computational and memory requirements limit its feasibility when dealing with large datasets. Another straightforward approach to combine the advantages of deep learning methods and the possibility of uncertainty modeling is the Bayesian Neural Network (BNN) (Goan and Fookes, 2020). Unlike standard neural networks that produce point estimates, BNNs assign probability distributions over the network parameters, thereby inducing distributions for outputs. However, BNNs reveal expressiveness limitations because the approximate inference may not reliably capture true posterior behavior (Foong et al., 2020). Therefore, the adaptation and application of DGMs are crucial for achieving rich representational capacity, flexible modeling frameworks, and the ability to effectively handle large-scale data.

Different DGMs have been investigated in system identification applications. VAEs are flexible to cooperate with time series modeling architectures such as Recurrent Neural Network (RNN) (Girin et al., 2021) and Nonlinear Autoregressive eXogenous (NARX) (Paniagua and López, 2024), which is beneficial in uncovering the time-dependent features from data. However, because the VAE models the output as a Gaussian characterized by means and variances, joint optimization of both mean and variance in the VAE leads to the well-known problem of shrinkage or underestimation of variance (Mattei and Frellsen, 2018). Unlike VAEs, NFs are restricted in their architectural choices due to the requirement of invertibility, which limits their direct applicability to time series modeling. On the other side, NFs provide exact and expressive density estimation capabilities, qualifying them for capturing uncertainty in data and enhancing the representational capacity of proba-

bilistic forecasts. Diffusion models release the architectural constraints and manifest the strength in generating high-quality and coherent predictions. Nonetheless, their iterative sampling procedure incurs substantial computational overhead. Balancing expressiveness, computational efficiency, and sample quality, ***normalizing flows*** are chosen in this work as the generative backbone of probabilistic system identification.

Based on the results of system identification, ***state estimation*** is the process of inferring the value of hidden, time-varying state variables of a dynamical system from a history of noisy and potentially partial observations and known control inputs. Because of the noise in the sensors (measurement noise) and unmodeled disturbances in the system itself (process noise), state estimation is naturally probabilistic. Thus, estimating the target states is computing the posterior distribution of target states. The traditional method for state estimation is the Luenberger State Observer, which assumes the systems are deterministic and both the system dynamics and measurement models are linear and time-invariant. Taking stochasticity into account, Kalman Filter is the most widely applied method. Beyond system linearity, a Kalman filter assumes the process and measurement noises are additive Gaussian noise with zero mean. In order to handle nonlinear systems, Extended Kalman Filter linearizes the nonlinear functions around the state estimate. However, the linearization introduces errors, leading to wrong estimation or even divergence. An improved modification is the Unscented Kalman Filter, which carefully selects the means and variances and propagates them through the nonlinear functions. Nevertheless, it is computationally inefficient and still only compatible with Gaussian noises.

Through relaxing the need for precise modeling of dynamical systems and avoiding restrictive assumptions about uncertainty–while preserving expressiveness, modeling flexibility, and scalability to large datasets–the chosen architecture for probabilistic system identification, NF, also provides a foundation for accurate, efficient and reliable probabilistic state estimation. From a data-centric perspective, parameter identification involves mapping time series to discrete data points; dynamics identification operates either between data points or across entire time series; and state estimation entails inferring data points or reconstructing time series from temporal data. In order to address these tasks, it is necessary to extend normalizing flows, given the limited research currently available in this area. Bridging this gap not only broadens the applicability of NFs but also contributes to advancing probabilistic inference techniques for complex temporal systems.

Having explored system identification and state estimation through probabilistic inference, we can advance toward data-driven control by leveraging data as a central source of insight into system behavior, enabling more adaptive, scalable, and robust control strategies. This is especially advantageous in situations where analytical models are incomplete, uncertain, or impractical to derive. One unmediated application is model-free control, which obtains control inputs directly from observations and reference trajectories. This

enables fast path planning without requiring an explicit system model. Furthermore, for more reliable and interpretable control, a preferable approach is to employ the identified dynamics within a Model Predictive Control (MPC) framework. This integration of learned models with numerical solvers and optimization routines allows for anticipatory decision-making based on predicted system behavior. Notably, in cases where simulation models are complex and the Jacobian of dynamics is analytically intractable, learning-based representations offer differentiable approximations that are inherently compatible with gradient-based solvers and optimizers. This compatibility facilitates efficient and scalable MPC implementations. Besides, the probabilistic nature of the extrated models enables uncertainty-aware control strategies, enhancing robustness and safety in dynamic and uncertain environments.

In summary, this work is motivated by three key considerations:

- Probabilistic inference provides a principled framework for modeling uncertainty and extracting knowledge from uncertainty, enabling system identification and state estimation based solely on observations in situations. This is particularly valuable in scenarios where first-principles models are unavailable, incomplete, or prohibitively complex.

- System identification and state estimation are required to be expressive for representing complex models, capable of modeling uncertainty, and scalable to large data sets. Deep Generative Models (DGMs), in particular Normalizing Flows (NFs), fulfill these criteria. However, existing architectures must be further developed to adapt to sequential data more flexibly.

- The results of probabilistic system identification can be integrated in Model Predictive Control (MPC), paving the way for interpretable, reliable, and robust control strategies. This fusion of learned models with optimization-based control frameworks holds promise for advancing autonomous decision-making in complex, uncertain environments.

2 Background and Preliminaries

This section introduces the necessary foundational concepts and methodologies for understanding the rest of this work. Before the main contents, we explain the unified notations applied in this whole work. We first review classical Bayesian inference methods. Then we introduce deep probabilistic models, including Deep Gaussian Process (DGP), Bayesian Neural Network (BNN), and Variational Autoencoder (VAE). Next, we present the core methods of this work, Normalizing Flows (NFs), a powerful class of generative models for flexible density estimation. Finally, we bridge probabilistic inference to control methods, covering system identification, state estimation, and model predictive control.

2.1 Notations

To maintain clarity throughout this document, we adopt the following notation:

Basic symbols

- Scalars are denoted by lowercase letters, e.g., x.
- Vectors are denoted by bold lowercase letters, e.g., $\boldsymbol{x}$.
- Matrices are denoted by bold uppercase letters, e.g., $\boldsymbol{X}$.
- Sets or collections of mixed or unspecified data types are denoted by calligraphic letters, e.g., $\mathcal{D}$.

Random variables

- Random variables may be scalars or vectors.
- Stochastic processes are denoted by italic capital letters with a time argument, e.g., $X(t)$, where t indicates time dependence.

Time series

- A time series of length T is denoted by $\boldsymbol{x}_{1:T}$, where the subscript indicates the time index.
- The i-th dimension of a time series is denoted by a superscript, e.g., $\boldsymbol{x}[i]$.

Variable modifications

- Estimated variables are denoted with a hat, e.g., $\hat{\boldsymbol{x}}$.

- Processed or transformed variables are denoted with a tilde, e.g., $\tilde{\boldsymbol{x}}$.

Probability and statistics

- Probability Density Function (PDF): $p(\cdot)$.
- Joint PDF: $p(\cdot, \cdot)$.
- Conditional PDF: $p(\cdot \mid \cdot)$.
- Expectation of a function $f(x)$ under PDF $p(x)$: $\mathbb{E}_{p(x)}[f(x)]$.
- Covariance between random vectors $\boldsymbol{x}$ and $\boldsymbol{y}$: $\mathrm{Cov}[\boldsymbol{x}, \boldsymbol{y}]$.
- Variance of a random variable $\boldsymbol{x}$: $\mathrm{Var}[\boldsymbol{x}]$.

Matrix operations

- Identity matrix: $\boldsymbol{I}$.
- Jacobian matrix: $\boldsymbol{J}$.
- Determinant of $\boldsymbol{J}$: $\det|\boldsymbol{J}|$.
- Trace of a matrix $\boldsymbol{X}$: $\mathrm{tr}(\boldsymbol{X})$.
- Transpose of a matrix $\boldsymbol{X}$: $\boldsymbol{X}^\top$.

2.2 Bayesian Inference

This chapter refers to the book *Bayesian Data Analysis* (Gelman et al., 2013), *Handbook of MCMC* (Allen, 2007).

2.2.1 Bayes' rule

Bayes' rule is derived from the chain rule of conditional probability. For target variables $\boldsymbol{\theta}$ and the corresponding observed data $\mathcal{D}$, the chain rule states that:

$$p(\boldsymbol{\theta}, \mathcal{D}) = p(\boldsymbol{\theta}|\mathcal{D})p(\mathcal{D}) = p(\mathcal{D}|\boldsymbol{\theta})p(\boldsymbol{\theta}). \tag{I.2}$$

Rearranging the terms gives Bayes' rule:

$$p(\boldsymbol{\theta}|\mathcal{D}) = \frac{p(\mathcal{D}|\boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\mathcal{D})}. \tag{I.3}$$

In Bayes' theorem, $p(\boldsymbol{\theta})$ is the initial distribution of the variables before any data samples are considered, which is called the prior distribution. $p(\boldsymbol{\theta}|\mathcal{D})$ is the posterior distribution of

the target parameter after being updated by the observed data. $p(\mathcal{D}|\boldsymbol{\theta})$ is the likelihood of expected data given the setting of parameters $\boldsymbol{\theta}$. $p(\mathcal{D})$ is called the marginal likelihood or evidence of expected data across all possible parameter settings, i.e., $\Sigma_{\boldsymbol{\theta}'} p(D|\boldsymbol{\theta}')p(\boldsymbol{\theta}')\Delta\boldsymbol{\theta}'$ or $\int p(D|\boldsymbol{\theta}')p(\boldsymbol{\theta}')d\boldsymbol{\theta}'$ for continuous $\boldsymbol{\theta}$. In practice, it is often treated as a normalizing constant and ignored. The unnormalized posterior is then given by:

$$p_u(\boldsymbol{\theta}|\mathcal{D}) \propto p(\mathcal{D}|\boldsymbol{\theta})p(\boldsymbol{\theta}). \tag{I.4}$$

Normally, the posterior distribution does not have a closed-form expression. Instead, we approximate it using numerical methods. Computational algorithms that rely on repeated random sampling to obtain numerical results refers to "Monte Carlo" methods. Monte Carlo methods provide ways to approximate the posterior distribution by expectations:

$$E = \mathbb{E}_{p(\boldsymbol{\theta}|\mathcal{D})}[f(\boldsymbol{\theta})] = \int f(\boldsymbol{\theta})p(\boldsymbol{\theta}|\mathcal{D})d\boldsymbol{\theta}, \tag{I.5}$$

where $f(\boldsymbol{\theta})$ is any function of interest, for example, $f(\boldsymbol{\theta}) = \boldsymbol{\theta}$. As the simplest way, we can approximate it using standard numerical techniques such as a Riemann sum over a discrete grid of points:

$$E \approx \Sigma_{\boldsymbol{\theta}'} f(\boldsymbol{\theta}')\Delta\boldsymbol{\theta}'. \tag{I.6}$$

However, this approach is only feasible for low-dimensional $\boldsymbol{\theta}$ due to the curse of dimensionality. Besides, for complex distributions, the discretization is highly inefficient. Therefore, for high-dimensional parameters and complex distributions, we often resort to sampling-based or optimization-based methods to approximate the posterior distribution, which will be discussed in the following sections.

2.2.2 Markov Chain Monte Carlo

The "Markov Chain" aspect indicates that these algorithms generate a sequence of samples where each sample depends only on the previous one. In contrast to the basic Monte Carlo methods, Markov Chain Monte Carlo (MCMC) methods produce auto-correlated samples, so that each new sample adds less new information than an independent draw would. Different sequences updating algorithms lead to different MCMC methods. The goal of MCMC is to construct a Markov chain that has the desired posterior distribution as its equilibrium distribution. After running the chain for a sufficient number of iterations, the samples generated from the chain can be treated as samples from the target posterior distribution.

Metropolis-Hastings (MH) algorithm

The MH algorithm, which is named after Metropolis (1953) and Hastings (1970), is a widely used MCMC method. The steps of the MH algorithm are as follows:

1. Initialize the chain with a starting point $\boldsymbol{\theta}_{(0)}$, choose a proposal distribution $q(\boldsymbol{\theta}'|\boldsymbol{\theta})$.
2. For each iteration i:
   (a) Propose a new state $\boldsymbol{\theta}'_{(i)}$ from the proposal distribution $q(\boldsymbol{\theta}'_{(i)}|\boldsymbol{\theta}_{(i-1)})$.
   (b) Calculate the transition probability, also known as the acceptance ratio:

$$T(\boldsymbol{\theta}'_{(i)}|\boldsymbol{\theta}_{(i-1)}) = \min\{1, \frac{p(\boldsymbol{\theta}'_{(i)}|\mathcal{D})q(\boldsymbol{\theta}_{(i-1)}|\boldsymbol{\theta}'_{(i)})}{p(\boldsymbol{\theta}_{(i-1)}|\mathcal{D})q(\boldsymbol{\theta}'_{(i)}|\boldsymbol{\theta}_{(i-1)})}\}. \tag{I.7}$$

   (c) Generate a uniform random number $u \sim U(0,1)$. If $u < T(\boldsymbol{\theta}'_{(i)}|\boldsymbol{\theta}_{(i-1)})$, accept the proposed state and set $\boldsymbol{\theta}_{(i)} \leftarrow \boldsymbol{\theta}'_{(i)}$; otherwise, reject it and set $\boldsymbol{\theta}_{(i)} \leftarrow \boldsymbol{\theta}_{(i-1)}$.
3. Repeat step 2 for a large number of iterations to obtain a sequence of samples $\{\boldsymbol{\theta}_{(1)}, \boldsymbol{\theta}_{(2)}, \ldots, \boldsymbol{\theta}_{(N)}\}$.

From the sequence of samples, we can contruct a Monte Carlo estimator:

$$p(\boldsymbol{\theta}|\mathcal{D}) \approx \frac{1}{N}\Sigma_{i=1}^{N}\delta_{\boldsymbol{\theta}_{(n)}}(\boldsymbol{\theta}). \tag{I.8}$$

Although this procedure allows flexible approximation of complex posterior distribution, it suffers from slow convergence rate due to its random exploration nature. Thus, gradient-based MCMC methods are proposed to incorporate the structure of the target density, leading to faster sampling.

Metropolis-adjusted Langevin algorithm (MALA)

The MALA (Besag, 2004) combines the MH algorithm with Langevin diffusion, whose stationary distribution is the target density. In general, for a stochastic process $\Theta(t)$, the Langevin diffusion is described by the following stochastic differential equation (SDE):

$$d\Theta(t) = \frac{1}{2}\nabla log\ p(\Theta(t))dt + dW(t), \tag{I.9}$$

where $W(t)$ denotes a standard Wiener process (Gabbiani and Cox, 2010). The Wiener process is a continuous-time stochastic process with independent, normally distributed increments, which can be interpreted as Brownian motion. It has the following properties:

(1) $W(0) = \mathbf{0}$;

(2) for $t > s \geq 0$, $W(t) - W(s) \sim \sqrt{t-s}\,\mathcal{N}(\mathbf{0}, \boldsymbol{I})$;

(3) for $t_1 > s_1 \geq 0$, and $t_2 > s_2 \geq 0$, $W(t_1) - W(s_1)$ and $W(t_2) - W(s_2)$ are independent.

The MALA algorithm uses the first order Euler discretization of the SDE, resulting in the following proposal distribution:

$$\Theta(t) = \Theta(i-1) + \frac{\epsilon^2}{2}\nabla \log p(\Theta(i-1)) + \epsilon\,\mathcal{N}(0, \boldsymbol{I}), \tag{I.10}$$

where ϵ is a step size parameter. Because of the discretization, convergence to the target density is no longer guaranteed. Therefore, the MH algorithm is applied to correct the bias introduced by the discretization. For each iteration i, the MALA algorithm updates the state as follows:

(a) Compute the gradient of the log-posterior at the current state: $\nabla \log p(\boldsymbol{\theta}_{(i-1)}|\mathcal{D})$.

(b) Propose a new state $\boldsymbol{\theta}'_{(i)}$ using the Langevin dynamics I.10:

$$\boldsymbol{\theta}'_{(i)} = \boldsymbol{\theta}_{(i-1)} + \frac{\varepsilon^2}{2}\nabla \log p(\boldsymbol{\theta}_{(i-1)}|\mathcal{D}) + \varepsilon\,\mathcal{N}(0, \boldsymbol{I}). \tag{I.11}$$

(c) Calculate the transition probability according to I.7. Specifically, the proposal distribution $q(\boldsymbol{\theta}'_{(i)}|\boldsymbol{\theta}_{(i-1)}) = \mathcal{N}(\boldsymbol{\theta}'|\mu(\boldsymbol{\theta}_{(i-1)}, \varepsilon), \varepsilon^2\boldsymbol{I})$, which is derived from the Langevin dynamics.

(d) (same as MCMC) Generate a uniform random number $u \sim U(0,1)$. If $u < T(\boldsymbol{\theta}'_{(i)}|\boldsymbol{\theta}_{(i-1)})$, accept the proposed state and set $\boldsymbol{\theta}_{(i)} \leftarrow \boldsymbol{\theta}'_{(i)}$; otherwise, reject it and set $\boldsymbol{\theta}_{(i)} \leftarrow \boldsymbol{\theta}_{(i-1)}$.

Compared to the MH, MALA converges faster and scales better in higher dimensions, at the cost of requiring gradients and tuning. As a more advanced MCMC method, Hamiltonian Monte Carlo (HMC) (Duane et al., 1987) further improves sampling efficiency by simulating the whole sampling trajectory as Hamiltonian dynamics.

2.2.3 Variational Inference

Different from MCMC, Variational Inference (VI) (Jordan et al., 1998; Wainwright and Jordan, 2008) is an optimization-based method that approximates the target posterior distribution $p(\boldsymbol{\theta}|\mathcal{D})$ with a simpler, parameterized distribution $q_{\boldsymbol{\phi}}(\boldsymbol{\theta})$, where $\boldsymbol{\phi}$ are the variational parameters. The goal of VI is to find the optimal variational parameters $\boldsymbol{\phi}^*$

that minimize the distance between the approximate distribution and the true posterior. The distance is normally measured by the Kullback-Leibler (KL)-divergence (Kullback and Leibler, 1951), which is defined as:

$$D_{\mathrm{KL}}(q(z)||p(z)) = \mathbb{E}_{q(z)}[\log \frac{q(z)}{p(z)}] = \int q(z) \log \frac{q(z)}{p(z)} dz. \tag{I.12}$$

Thus, the optimization problem of VI can be formulated as:

$$\boldsymbol{\phi}^* = \arg\min_{\boldsymbol{\phi}} D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\boldsymbol{\theta})||p(\boldsymbol{\theta}|\mathcal{D})). \tag{I.13}$$

Substituting Bayes' rule I.3 into KL-divergence, we have:

$$D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\boldsymbol{\theta})||p(\boldsymbol{\theta}|\mathcal{D})) = D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\boldsymbol{\theta})||\frac{p(\mathcal{D}|\boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\mathcal{D})}). \tag{I.14}$$

Same as the aforementioned "intractable posterior" problem, this divergence is directly intractable due to the presence of the evidence $p(\mathcal{D})$.

Expanding the above equation and extending it with the definition of KL-divergence, we have:

$$\begin{aligned} D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\boldsymbol{\theta})||p(\boldsymbol{\theta}|\mathcal{D})) &= \mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{\theta})}[\log q_{\boldsymbol{\phi}}(\boldsymbol{\theta}) - \log p(\mathcal{D}|\boldsymbol{\theta}) - \log p(\boldsymbol{\theta}) + \log p(\mathcal{D})] \\ &= \mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{\theta})} \log q_{\boldsymbol{\phi}}(\boldsymbol{\theta}) - \mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{\theta})} \log p(\mathcal{D}|\boldsymbol{\theta}) - \mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{\theta})} \log p(\boldsymbol{\theta}) + \mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{\theta})} \log p(\mathcal{D}). \end{aligned} \tag{I.15}$$

Since $p(\mathcal{D})$ is independent of the variational parameters $\boldsymbol{\phi}$, the last term is reduced to $\log p(\mathcal{D})$. Combining the terms $p(\mathcal{D}|\boldsymbol{\theta})p(\boldsymbol{\theta}) = p(\boldsymbol{\theta}, \mathcal{D})$ gives

$$D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\boldsymbol{\theta})||p(\boldsymbol{\theta}|\mathcal{D})) = \mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{\theta})} \log q_{\boldsymbol{\phi}}(\boldsymbol{\theta}) - \mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{\theta})} \log p(\boldsymbol{\theta}, \mathcal{D}) + \log p(\mathcal{D}) \tag{I.16}$$

Or combining the terms $q_{\boldsymbol{\phi}}(\boldsymbol{\theta})$ and $p(\boldsymbol{\theta})$ via KL-divergence gives:

$$D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\boldsymbol{\theta})||p(\boldsymbol{\theta}|\mathcal{D})) = D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\boldsymbol{\theta})||p(\boldsymbol{\theta})) - \mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{\theta})} \log p(\mathcal{D}|\boldsymbol{\theta}) + \log p(\mathcal{D}) \tag{I.17}$$

Rearranging the terms in Equation I.16 gives:

$$\log p(\mathcal{D}) = D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\boldsymbol{\theta})||p(\boldsymbol{\theta}|\mathcal{D})) + \mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{\theta})} \log p(\boldsymbol{\theta}, \mathcal{D}) - \mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{\theta})} \log q_{\boldsymbol{\phi}}(\boldsymbol{\theta}). \tag{I.18}$$

Since the KL-divergence is always non-negative, we have:

$$\log p(\mathcal{D}) \geq \mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{\theta})} \log p(\boldsymbol{\theta}, \mathcal{D}) - \mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{\theta})} \log q_{\boldsymbol{\phi}}(\boldsymbol{\theta}). \tag{I.19}$$

Because of this inequality, the right hind side is called the Evidence Lower Bound (ELBO), also known as the variational lower bound. Formulating the ELBO using Equation I.17 gives:

$$\text{ELBO} = \mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{\theta})} \log p(\mathcal{D}|\boldsymbol{\theta}) - D_{\text{KL}}(q_{\boldsymbol{\phi}}(\boldsymbol{\theta})||p(\boldsymbol{\theta})). \tag{I.20}$$

Then, minimizing the KL-divergence is equivalent to maximizing the ELBO, leading to the equivalent optimization problem:

$$\boldsymbol{\phi}^* = \arg\max_{\boldsymbol{\phi}} \text{ELBO} = \arg\min_{\boldsymbol{\phi}} [D_{\text{KL}}(q_{\boldsymbol{\phi}}(\boldsymbol{\theta})||p(\boldsymbol{\theta})) - \mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{\theta})} \log p(\mathcal{D}|\boldsymbol{\theta})]. \tag{I.21}$$

To complete the optimization, we need to choose a suitable family of variational distributions $q_{\boldsymbol{\phi}}(\boldsymbol{\theta})$. A common choice is the mean-field family (Bishop, 2006), which assumes that the variables $\boldsymbol{\theta}$ can be factorized into groups that are independent under q:

$$q_{\boldsymbol{\phi}}(\boldsymbol{\theta}) = \Pi_i q_{\boldsymbol{\phi}_i}(\boldsymbol{\theta}_i), \tag{I.22}$$

where each parameter $\boldsymbol{\phi}_i$ governs a subset of the variables $\boldsymbol{\theta}_i$. Then, the ELBO is iteratively optimized by updating each parameter while holding the others fixed. The solution of the optimal distribution for each group is given by:

$$\log q^*_{\boldsymbol{\phi}_i}(\boldsymbol{\theta}_i) = \mathbb{E}_{\Pi_{j\neq i} q_{\boldsymbol{\phi}_i}(\boldsymbol{\theta}_i)} [\log p(\boldsymbol{\theta}, \mathcal{D})], \tag{I.23}$$

where the expectation is taken with respect to the distributions of all other groups $j \neq i$. This algorithm is called coordinate ascent variational inference (CAVI) (Bishop, 2006).

This assumption simplifies the optimization problem, and thus makes it scalable for large dimensional variables and also enables parallelizable computation. However, it may also limit the expressiveness of the variational distribution, especially when the posterior correlations are crucial, leading to suboptimal approximations of the true posterior. Summarizing, VI is generally faster than MCMC and scales better to large datasets, but it may provide less accurate approximations of the posterior distribution (Blei et al., 2017).

2.3 Deep Probabilistic Models

Deep probabilistic models combine the representational power of deep learning with the principled uncertainty modeling of probabilistic inference methods. This integration aims to provide both high expressiveness and scalability to large datasets. In the following section, we briefly review three prominent deep probabilistic models: the Deep Gaussian Process, the Bayesian Neural Network, and the Variational Autoencoder.

2.3.1 Deep Gaussian Process

A Gaussian process is a stochastic process, where at each time point the random variables follow normal distributions. It can be interpreted as an infinite-dimensional Gaussian distribution. For an N-dimensional random variable $\boldsymbol{x} = [x_1, \ldots, x_N]^T$, a Gaussian process $GP(\boldsymbol{x})$ is fully specified by its mean function $\mu(\cdot)$ and covariance function $k(\cdot, \cdot)$:

$$\begin{bmatrix} GP(x_1) \\ \cdots \\ GP(x_N) \end{bmatrix} \sim \mathcal{N}(\begin{bmatrix} \mu(x_1) \\ \cdots \\ \mu(x_N) \end{bmatrix}, \begin{bmatrix} k(x_1, x_1), \ldots, k(x_1, x_N) \\ \cdots \\ k(x_N, x_1), \ldots, k(x_N, x_N) \end{bmatrix}) \tag{I.24}$$

Using a Gaussian process to infer variables from observed data based on prior distributions leads to Gaussian process regression (GPR) (Rasmussen and Williams, 2008). For a regression task with training data $\boldsymbol{x}$ and the ground truth $\boldsymbol{y}$, the GPR model is employed by a latent function $f(\cdot)$:

$$y_i = f(x_i) + \epsilon_i, \quad i = 1, \ldots, N, \tag{I.25}$$

where $\epsilon_i \sim \mathcal{N}(0, \sigma_\epsilon^2)$ are i.i.d. noises for all dimensions. So, the likelihood of $\boldsymbol{y}$ is given by:

$$p(\boldsymbol{y}|\boldsymbol{f}) = \prod_{i=1}^{N} p(y_i|f_i) = \prod_{i=1}^{N} \mathcal{N}(y_i|f_i, \sigma_\epsilon^2). \tag{I.26}$$

The prior distribution over functions $f(\cdot)$ is assumed to a zero-mean Gaussian process:

$$f(\cdot) \sim GP(0, k(\cdot, \cdot)). \tag{I.27}$$

The kernel function $k(\cdot, \cdot)$ is usually chosen to be the squared exponential (SE) kernel function, or as known as the Radial Basis Function (RBF), defined as:

$$k(x_i, x_j) = \sigma_{SE}^2 \exp(-\frac{1}{2\ell^2} \left\| x_i - x_j \right\|^2), \tag{I.28}$$

where σ_{SE} and $\ell >$ are vertical and horizontal scale parameters, which control the vertical span and smoothness of the function, respectively. We denote the collection of latent functions as $\boldsymbol{f} = [f(x_1), \ldots, f(x_N)]^T$. Then, the posterior distribution of $\boldsymbol{y}$ can be computed analytically using the marginal likelihood over $\boldsymbol{y}$:

$$p(\boldsymbol{y}|\boldsymbol{x}) = \int \prod_{i=1}^{N} p(y_i|f_i) p(f_i|x_i) d\boldsymbol{f} = \mathcal{N}(\boldsymbol{y}|\boldsymbol{0}, \boldsymbol{K} + \sigma_\epsilon^2 \mathbb{I}), \quad \boldsymbol{K} = k(\boldsymbol{x}, \boldsymbol{x}). \tag{I.29}$$

The hyperparameters $\boldsymbol{\phi} = \{\sigma_{SE}, \ell, \sigma_\epsilon\}$ can be optimized by maximizing the log marginal likelihood $log\ p(\boldsymbol{y}|\boldsymbol{x}, \boldsymbol{\phi})$.

The GPR is widely used for regression tasks owing to its non-parametric nature and its capacity to quantify uncertainty. However, its computational complexity scales cubically with the number of training points, i.e., $\mathcal{O}(N^3)$ for an N-dimensional input. Additionally, calculating the Gaussian processes requires the entire storage of training datasets, incurring a memory cost of $\mathcal{O}(M^2)$, which grows quadratically with the dataset size M. These limitations hinder its scalability to high-dimensional inputs and large datasets. In addition, the expressiveness of GPR is constrained by the choice of kernel function and its inability to perform hierarchical feature extraction.

To improve expressiveness, DGP (Damianou and Lawrence, 2013) is proposed as multilayer hierarchical generalizations of Gaussian processes. The input $\boldsymbol{x}$ is passed through multiple layers of Gaussian processes to obtain the output $\boldsymbol{y}$. For an L-layer DGP, the model extends the Gaussian process I.25 by:

$$y_i = f_L(f_{L-1}(\dots f_1(x_i) + \epsilon_{i,1} \dots)) + \epsilon_L,\ f_l(\cdot) \sim GP(0, k_l(\cdot,\cdot)),\ l = 1, \dots, L,\ i = 1, \dots, n. \tag{I.30}$$

The deep architecture enables Gaussian processes to learn more complex data patterns and hierarchical representations. It can be seen as a deep neural network with infinitely wide layers (Neal, 1996), where each layer applies a Gaussian process, instead of nonlinear matrix transformation. However, due to the computational cost and storage consumption the DGP is still cumbersome to apply in practice.

2.3.2 Bayesian Neural Networks

Another approach to constructing deep probabilistic models involves integrating Bayesian inference with deep neural networks. A Bayesian Neural Network (BNN) (Neal, 1996; MacKay, 1995) places prior distributions over the network's weights and biases, allowing uncertainty to be modeled directly within the architecture.

The goal of training a BNN is to infer the posterior distribution over the parameters $\boldsymbol{\theta} = [\boldsymbol{w}, \boldsymbol{b}]$ given the training data $\mathcal{D}$, i.e., $p(\boldsymbol{\theta}|\mathcal{D})$. VI is commonly employed to approximate the posterior by optimizing the ELBO I.20. To facilitate gradient-based optimization via backpropagation–similar to standard neural networks–the reparameterization trick (Kingma and Welling, 2022) is used. This technique expresses random variables as deterministic variables of noise, enabling efficient and differentiable sampling during training. For example, for a Gaussian variational distribution $q_{\boldsymbol{\phi}}(\boldsymbol{\theta}) = \mathcal{N}(\boldsymbol{\theta}|\boldsymbol{\mu}, \mathrm{diag}(\boldsymbol{\sigma}^2))$, we can reparameterize $\boldsymbol{\theta}$ as:

$$\boldsymbol{\theta} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}), \tag{I.31}$$

where $\odot$ represents the Hadamard product, i.e., the element-wise product, and the variational parameters here are $\boldsymbol{\phi} = [\boldsymbol{\mu}, \boldsymbol{\sigma}]$. To train a BNN on large datasets, Stochastic

Gradient Descent (SGD) (Bottou, 1999) is typically employed to optimize the ELBO over mini-batches. Unlike standard gradient descent, SGD estimates the gradient at each iteration using a randomly selected subset of the data. Similar to Equation I.21, the variaitonal objective of a BNN becomes:

$$\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\phi}) = \frac{1}{M} \sum_{i=1}^{M} [D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\boldsymbol{\theta}) || p(\boldsymbol{\theta})) - \mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{\theta})} \log p(\mathcal{D}_i | \boldsymbol{\theta})], \tag{I.32}$$

where each mini-batch $\mathcal{D}_i \subset \mathcal{D}$ is drawn from the equally sized subsets of the training data $\mathcal{D} = \{\mathcal{D}_1, \ldots, \mathcal{D}_M\}$. During each optimization step, the auxiliary noise $\boldsymbol{\epsilon}$ is sampled multiple times from $\mathcal{N}(\mathbf{0}, \boldsymbol{I})$ to perform Monte Carlo estimation of the expectation terms in the loss function $\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\phi})$. The gradient $\nabla_{\boldsymbol{\phi}} \mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\phi})$ is then computed via backpropagation and averaged across all samples. Finally, the variational parameter $\boldsymbol{\phi}$ is updated by

$$\boldsymbol{\phi} \leftarrow \boldsymbol{\phi} - \alpha \nabla_{\boldsymbol{\phi}} \mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\phi}), \tag{I.33}$$

where α is the learning rate. This procedure is called Bayes by Backprop (BbB) (Blundell et al., 2015).

In comparison to DGP, the BNN demonstrates better scalability to large datasets and improved representational capacity. Moreover, the BbB method facilitates the compatibility of BNN with deep learning techniques and thus enables flexible implementation in different neural network architectures, such as Convolutional Neural Network (CNN) (Shridhar et al., 2019), RNN (Fortunato et al., 2019), and Transformer (Müller et al., 2024). Nevertheless, several works raise concerns with BNNs including wrong uncertainty estimation in between separated regions of obsertations (Foong et al., 2019, 2020), poor generalization ability under domain shift (Izmailov et al., 2021), and sensitive modeling ability to the choice of prior distributions (Silvestro and Andermann, 2020). These limitations motivate the applications of more powerful generative models, such as variational autoencoders and normalizing flows, which are discussed in the following sections.

2.3.3 Variational Autoencoder

In classical VI methods, the mapping from the parameters of interest $\boldsymbol{\theta}$ to the observations $\mathcal{D}$ is assumed to be known in advance. This is called a fully-observable model. and the approximate probabilistic models are parameterized by $\boldsymbol{\phi}$. In contrast, a Variational Autoencoder (VAE) is constructed with deterministic neural networks and introduces variability with latent variables. Instead of directly approximating the posterior distribution over the parameters using VI, Variational Autoencoder (VAE) (Kingma and Welling, 2022; Rezende et al., 2014) introduces a set of latent variables $\boldsymbol{z}$ to model the data generation

process. The generative model is defined as:

2.4 Normalizing Flows

2.4.1 Basics

2.4.2 Discrete Normalizing Flows

Coupling layers

Autoregressive flows

2.4.3 Continuous Normalizing Flows

Neural ODEs

FFJORD

2.5 BayesFlow

2.6 From Inference to Control

2.6.1 Bayesian System Identification

2.6.2 Bayesian State Estimation

2.6.3 Model-free Control

2.6.4 Model Predictive Control

3 Literature Review

This section reviews the relevant literature in three aspects: applications of deep probabilistic models especially normalizing flows for Bayesian inference; state-of-the-art deep probabilistic methods for system identification and state estimation; and data-driven control methods based on deep probabilistic models.

3.1 Deep Probabilistic Models for Bayesian Inference

3.2 Deep Probabilistic Models for System Identification and State Estimation

3.3 Data-driven Control based on Deep Probabilistic Models

4 Objectives and Contributions

Starting from BayesFlow, we explore the potential of normalizing flows for parameter identification, system identification, and state estimation. Moreover, we aim to bridge the gap between probabilistic inference and control tasks. The primary objectives of this dissertation are as follows:

- To investigate the capabilities of normalizing flows for system parameter identification.
- To extend BayesFlow for paramter identification to perform system identification and state estimation tasks.
- To assess the performance of the proposed models on benchmark datasets and real-world scenarios.
- To analyze the interpretability and uncertainty quantification capabilities of the learned models.
- To examine the applicability of the identified systems in control tasks.

The key contributions of this dissertation can be summarized as:

- Enhancement of the original BayesFlow framework to improve robustness, enabling its deployment in real-world applications.
- Integration of BayesFlow with sensitivity analysis to facilitate the estimation of parameters with different degrees of identifiability.
- Development of Stochastic BayesFlow by combining stochastic normalizing flows with BayesFlow, aimed at robust parameter identification under uncertainty.
- Extension of BayesFlow to learn sequence-to-sequence models for system identification and state estimation.
- Proposal of FlowSDE, a new method for learning unknown stochastic dynamical systems by embedding normalizing flows within stochastic differential equations.
- Investigation of the compatibility of normalizing flow-based models within control frameworks.

5 Outline

The remainder of this thesis is organized as follows:

- Chapter II outlines the methodologies developed and employed throughout this work. These methods are grouped into three categories: parameter identification, dynamics identification, and state estimation. Finally, data-driven control strategies based on dynamics identification approaches are introduced.
- Chapter III presents the experimental validation of the proposed methods, structured according to the aforementioned categories.
- Chapter IV summarizes the key findings of the dissertation and proposes directions for future research.

Chapter II

Methodologies

1 Parameter Identification

BayesFlow
- Stochastic BayesFlow
- BayesFlow for differently identifiable parameters

2 Dynamics Identification

IMFlow
- FMFlow
- FlowSDE

3 State Estimation

IMFlow ability limited

limitation for bayesflow for identifying highly dynamical data

transformer-bayesian neural network better for identification of battery state-of-health

Luenberger Observer with FlowSDE

4 Data-driven Control

Model free control with IMFlow
- Model predictive control with FMFlow
- Model predictive control with FlowSDE

Chapter III

Experiments and Results

1 Overview

2 Parameter Identification

von der pol single track car quadcopter

3 Dynamics Identification

synthetic data physics-based real-world

4 State Estimation

5 Data-driven Control

model-free control
    model predictive control

Chapter IV

Discussion and Conclusion

Chapter V

Abstract

Chapter VI

Zusammenfassung

Own Contributions

This chapter gives an overview of my contributions in distinction to team efforts.

Own share in data acquisition and data analysis

Own share in Data Acquisition.

Own share in Data Analysis.

Own Publications

In this section, all publications that I was a part of and contributed to during my Ph.D. work are listed. It is subdivided into *First Authorships* and *Co-Authorships*.

First Authorships - Peer Reviewed International Journal Publications

Y. Zhang, L. Mikelsons. **Advanced Modeling and Simulation in Engineering and Science**. *Sensitivity-guided iterative parameter identification and data generation with BayesFlow and PELS-VAE for model calibration*. doi:10.1186/s40323-023-00246-y.

First Authorships - Peer Reviewed International Conference Publications

Y. Zhang, L. Mikelsons. **2023 IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM)**. *Solving Stochastic Inverse Problems with Stochastic BayesFlow*. doi:10.1109/AIM46323.2023.10196190.

Y. Zhang, L. Mikelsons. **2024 IEEE 63rd Conference on Decision and Control(CDC)**. *Inverse Modeling with Conditional Normalizing Flows for Data-Driven Model Predictive*

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Co-Authorships

M.E. Heringhaus, Y. Zhang, A. Zimmermann, L. Mikelsons. **Sensors 2022**. *Towards Reliable Parameter Extraction in MEMS Final Module Testing Using Bayesian Inference*. doi:10.3390/s22145408.

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- Support in the development of a prediction model for process-reliable part removal on a fully automated laser cutting machine
- Further development and implementation of a novel computer vision approach for teaching a finite element structure simulation through U-Net architectures, with Keras
- Evaluation and training of the models as well as realization and identification of improvement potentials to increase model accuracy

Publications

Zhang, Y. and Mikelsons, L., IMFlow: Inverse Modeling with Conditional Normalizing Flows for Data-Driven Model Predictive Control, 2024 IEEE 63rd Conference on Decision and Control (CDC), Milan, Italy, 2024, pp. 3353-3359, doi: 10.1109/CDC56724.2024.10886862.
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MATLAB/Simulink, C++

Languages

English: Professional Proficiency
German: Professional Proficiency
Chinese: Native Proficiency

Technical Skills

Machine Learning/Deep Learning
Reinforcement Learning
Mathematics & Statistics
Data processing & -analysis
ML-Frameworks: PyTorch, Jax, TensorFlow/Keras

Problem-solving expertise

Implementation and optimization of algorithms
Interdisciplinary work
Creativity & critical thinking
Ability to think abstractly

Soft Skills

Explanation of complex concepts
Collaboration with researchers and industry partners
Independence & personal responsibility
Time management & prioritisation
Teamwork & collaboration
Openness to feedback
Curiosity & lifelong learning
Perseverance

Third-party funded projects

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Appendix

Acknowledgement

Eidesstattliche Versicherung

Statutory Declaration

1. Bei der eingereichten Dissertation zu dem Thema ***Probabilistic System Identification and State Estimation via Normalizing Flows*** handelt es sich um meine eigenständig erbrachte Leistung.
   *I herewith formally declare that I have written the submitted dissertation **Probabilistic System Identification and State Estimation via Normalizing Flows** independently.*
2. Ich habe nur die angegebenen Quellen und Hilfsmittel benutzt und mich keiner unzulässigen Hilfe Dritter bedient. Insbesondere habe ich wörtlich oder sinngemäß aus anderen Werken übernommene Inhalte als solche kenntlich gemacht.
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   I am aware of the legal consequences of this declaration. To the best of my knowledge I have told the pure truth and not concealed anything.

Augsburg, 05.09.2025 _____

Angabe zu verwendeten KI-basierter Elektronischer Hilfsmittel

Information on the use of AI-based tools

Zur Dokumentation der verwendeten Hilfsmittel ist der schriftlichen Ausarbeitung ein besonderer Anhang hinzugefügt, der eine Liste und Beschreibung aller verwendeter KI-basierter Hilfsmittel enthält. Der besondere Anhang zur Dokumentation der verwendeten Hilfsmittel erfüllt folgende Kriterien:

To document the tools used, a special appendix has been added to the written report, which contains a list and description of all AI-based tools used. The special appendix to document the tools used meets the following criteria:

1. Auflistung der Ziele, für die die KI-basierten Hilfsmittel in der vorliegenden Arbeit eingesetzt wurden.
   List of the goals for which the AI-based tools were used in this work.

2. Dokumentation der Verwendungsweise der KI-basierten Hilfsmittel.
   Documentation of how AI-based tools are used.

3. Nennung der Kapitel und Abschnitte der vorliegenden Arbeit, in denen die KI-basierten Hilfsmittel eingesetzt wurden, um Inhalte zu erzeugen.
   Identification of the chapters and sections of this work in which AI-based tools were used to generate content.

Der Gebrauch dieser Hilfsmittel inklusive Art, Ziel und Umfang des Gebrauchs wurde mit meinem offiziellen Betreuer **Prof.Dr.Ing. Lars Mikelsons** abgesprochen.

*The use of these aids, including the type, purpose and extent of use, has been agreed with my official supervisor **Prof.Dr.Eng. Lars Mikelsons**.*

Mir ist bewusst, dass insbesondere der Versuch einer nicht dokumentierten Nutzung KI- basierter Hilfsmittel als Täuschungsversuch zu werten ist:

I am aware that in particular the attempt to use AI-based tools without documentation is to be regarded as an attempt at deception:

Gem. § 16 Abs. 2 der Promotionsordnung „Dr. med. / dent.“: „Ergibt sich vor Aushändigung der Promotionsurkunde, dass der Kandidat/die Kandidatin bei einer Promotionsleistung getäuscht hat, so können einzelne oder alle Promotionsleistungen für ungültig erklärt werden. In schweren Fällen kann die Zulassung zum Promotionsverfahren zurückgenommen werden.“

According to § 16 paragraph 2 of the doctoral regulations for "Dr. med. / dent.": "If it emerges before the doctoral certificate is issued that the candidate has cheated in a doctoral achievement, individual or all doctoral achievements can be declared invalid. In serious cases, admission to the doctoral process can be withdrawn.

Und § 16 Abs. 2 der Promotionsordnung „Dr. sc. hum.“: „Ergibt sich vor Aushändigung der Promotionsurkunde, dass der Doktorand / Doktorandin bei einer Promotionsleistung getäuscht hat, so kann der Promotionsausschuss diese Promotionsleistung oder alle bisher erbrachten Promotionsleistungen für ungültig erklären. In besonders schweren Fällen kann der Promotionsausschuss die Annahme als Doktorand / Doktorandin endgültig widerrufen.“

And § 16 paragraph 2 of the doctoral regulations “Dr. sc. hum.”: “If it emerges before the doctoral certificate is issued that the doctoral candidate has cheated in a doctoral achievement, the doctoral committee can declare this doctoral achievement or all doctoral achievements to date invalid. In particularly serious cases, the doctoral committee can permanently revoke the acceptance as a doctoral candidate.”

Augsburg, 05.09.2025 _____